BPME: A Blind Perceptual Metric-Driven Method for Enhancing Degraded Polar Optical Remote Sensing Imagery

Zijun Wei, Chaozhen Lan, Fushan Yao, Longhao Wang, Tian Gao, and Hanyang Yu

***Abstract*—Optical remote sensing imagery is a vital data source for large-scale monitoring in polar regions. However, harsh imaging conditions often result in images with texture degradation. This imagery quality dilemma severely constrains the reliability of downstream analysis. Due to the uniqueness of the polar environment, current enhancement methods tend to introduce artifacts or suffer from limited domain generalization. To address these challenges, we propose a blind perceptual metric-driven enhancement (BPME) method for degraded polar optical imagery. Specifically, we first design a zero-reference metric system to precisely quantify degradation patterns. Subsequently, we construct the metric feature space to derive a blind perceptual metric prior (BPM-P) via clustering-based mapping. Finally, guided by the proposed BPM-P, a selective mechanism is employed to improve texture information adaptively. Comprehensive experiments confirm that the BPME outperforms state-of-the-art algorithms, as indicated by both qualitative and quantitative evaluations. The method also exhibits outstanding performance in downstream tasks such as contour extraction and ice flow velocity tracking. As a physically interpretable framework, BPME provides a robust solution for image degradation in polar remote sensing, effectively enhancing the interpretability and usability of optical remote sensing imagery.**

***Index Terms*—Blind perceptual metric prior (BPM-P), image degradation, polar remote sensing, texture enhancement, zero-reference metric.**

I. Introduction

DYNAMIC changes in polar regions serve as significant indicators of the global climate system. Satellite remote sensing provides a continuous and cost-effective means for monitoring these inaccessible regions [1]. When monitoring landform evolution and seasonal incidents during the melt period, the intuitive texture details in optical imagery offer an irreplaceable information dimension that SAR data struggles to obtain. To effectively capture the precise spatial patterns of surface texture, high-resolution panchromatic bands are often regarded as the optimal data source.

Nevertheless, polar panchromatic imagery frequently suffers from texture degradation, primarily manifesting in three typical scenarios as illustrated in Fig. 1. First, ice landscapes possess high uniform albedo properties, directly resulting in weak gradient information characterizing texture. Simultaneously, the frequent occurrence of cloud and fog conditions inevitably weakens texture intensity through strong scattering effects. Furthermore, the low solar elevation angle during polar imaging generally leads to gradient compression. Degraded texture signals emerge that are difficult to extract due to low contrast or are easily overwhelmed by noise. This effect makes it hard to get fine-scale surface measurements and keep an eye on dynamic processes, which is a major information bottleneck.

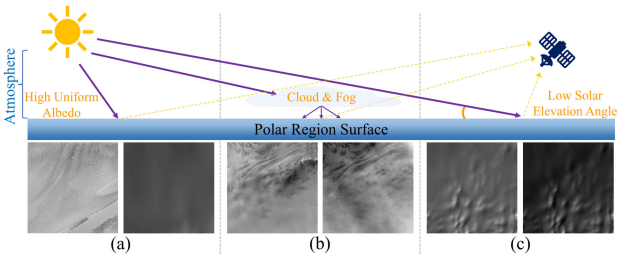


Fig. 1. Polar optical remote sensing image degradation scenarios. The challenge primarily falls into three main categories. (a) Inherent weak texture. (b) Atmospheric meteorological obscuration. (c) Complex illumination interference.

Texture enhancement is the principal solution at present [2], [3]. Extensive research has been conducted by scholars based on image quality optimization and visual studies [4], [5], [6]. The achievements encompass specialized methods for common challenges such as dehazing [7], [8] and low-light enhancement [9], [10]. They aim to provide a reliable data foundation for downstream tasks, including change detection and semantic segmentation [11], [12], [13], [14]. Despite this progress, these techniques often fall short in polar imagery for three main reasons.

1) The distinct physical conditions of polar regions differ significantly from those in typical low-latitude scenes, rendering many model assumptions [15], [16] ineffective when directly applied.
2) Frequent cloud and fog in polar areas make acquiring ideal ground-truth (GT) imagery exceptionally difficult, thereby limiting dataset construction [17], [18] and hindering the transferability of supervised deep learning models [19], [20].
3) Due to the complex and spatially heterogeneous nature of degradation in polar imagery, global processing approaches carry a high risk of amplifying noise and losing details [21], [22].

Received 3 September 2025; revised 9 December 2025; accepted 15 December 2025. Date of publication 19 December 2025; date of current version 7 January 2026. *(Corresponding author: Chaozhen Lan.)*

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Digital Object Identifier 10.1109/TGRS.2025.3646447

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To address these challenges, we propose a blind perceptual metric-driven enhancement (BPME) method. The term "blind" signifies that the enhancement process of the method requires no labeled datasets or reference knowledge to guide it. Compared to unsupervised techniques that similarly relax data requirements, BPME is driven by intrinsic image properties, further avoiding "black-box" model interventions and rigid physical assumptions. It demonstrates superior adaptability and mathematical interpretability in polar image processing. The achievement of BPME is to improve the effectiveness of visual interpretation and quantitative analysis based on degraded imagery. It possesses the capacity to enhance texture details while avoiding the introduction of noise and artifacts. This article presents the following main contributions.

1) We design a zero-reference metric system to precisely quantify heterogeneous degraded features, without relying on subjective human experience.
2) We construct a blind perceptual metric prior (BPM-P) via metric clustering-based mapping to provide spatially explicit guidance for enhancement, without requiring auxiliary data.
3) We propose a selective mechanism guided by BPM-P, designed for the enhancement of valid texture information. Extensive experimentation demonstrates that our method robustly outperforms state-of-the-art algorithms in various degradation scenarios.

This article organizes its parts as follows. Section II reviews related work and analyzes the limitations of existing methods in polar applications. Section III introduces the proposed method, including the zero-reference metric system (Section III-A), the generation of the BPM-P (Section III-B), and the prior-guided selective enhancement (Section III-C). Section IV details the experimental results and a rigorous evaluation of state-of-the-art methods. The article concludes in Section V.

II. Related Work

Considering the unique data characteristics and technical demands of the polar image enhancement, this section reviews related work from three primary paradigms. These paradigms provide a methodological overview centered on model interpretability, dataset requirements, and scenario adaptability, respectively.

A. Physical Model-Based Methods

Physical model-based methods tend to restore degraded images by inverting imaging models. Their critical model parameters are typically estimated on the basis of artificially designed assumptions. Theoretical foundations can divide related methodologies into two categories. The first one is based on the atmospheric scattering model. Among these, the dark channel prior (DCP) [23] proposed by He et al. for dehazing represents a landmark achievement. Subsequent research enhanced its universality by introducing more flexible priors, such as the generalized DCP (GDCP) [24] and saturation line prior (SLP) [25]. The second category relies on the Retinex theory. Well-known algorithms like LIME [26] and DUAL [27] primarily enhance degraded details by optimizing illumination estimation. As it is based on physical principles, interpretability is a significant advantage of this paradigm. Yet its assumptions mostly pertain to the general conditions at low latitudes. Owing to the distinctive variances contrasted with polar regions, existing models may lead to noise amplification and data distortion.

B. Zero-Reference Deep Learning Networks

Zero-reference deep learning networks restore single degraded images via designed loss optimization. "Zero-shot" approaches represent a specific framework within this paradigm. Relevant to polar degradation scenarios, this section highlights the dehazing and low-light enhancement algorithms. For dehazing, Li et al. [28] proposed zero-shot image dehazing (ZID), a method based on layer decoupling. This work was pioneering in addressing the reliance of data-driven methods on paired training data. Wei et al. [29] innovatively trained a zero-shot dehazing network by designing a redegraded model. Their subsequent work further advanced the field by combining a prior-based module with a convolutional neural network [30]. For low-light enhancement, Zero-DCE [31] and RUAS [32] are classic examples. Later studies incorporated theoretical knowledge, like camera response models, to further improve algorithmic rigor [33], [34]. The ZERO-IG, advanced by Shi et al. [35], employed an illumination-guided technique that achieved denoising and enhancement concurrently. Other low-light enhancement approaches have leveraged pretrained latent diffusion models [36], [37] or semantic guidance [38] to achieve optimization. This paradigm exemplifies remarkable adaptability in contexts characterized by data scarcity, as it does not necessitate the utilization of paired data or rigid physical assumptions. Nevertheless, given that network performance is predominantly influenced by the design of loss functions, the efficacy of this paradigm in specific applications such as polar image processing remains to be substantiated.

C. Feature-Based Methods

Feature-based methods are primarily categorized into contrast enhancement and high-frequency signal processing. Histogram equalization methods [39] are representative achievements in the former. While simple and effective, their global processing may introduce loss of details. The Wallis filter exhibits improved performance [40], owing to its local flexibility. High-frequency signal processing algorithms, such as the high-pass filter [41] and edge detection operators [42], have been incorporated into polar image processing workflows. However, the common challenge in these methods is distinguishing texture from noise. To mitigate this, GIV [43] introduced a nearly anisotropic orientation filter (NAOF), which selectively enhances textures via multidirectional convolution. But the performance of the classical methods is restricted to manual parameters. To overcome this limitation, subsequent initiatives have focused on incorporating more sophisticated intrinsic priors to improve performance. For instance, the advances have introduced the framelet transform with sparse regularization [44], spatially adaptive gamma

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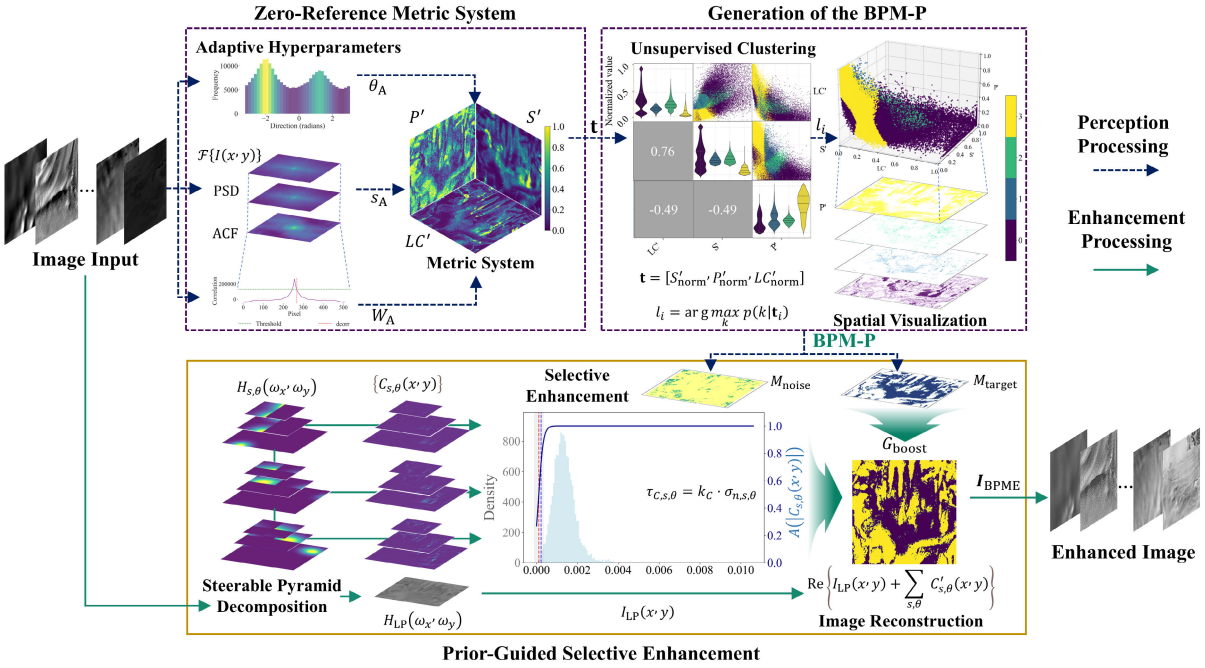


Fig. 2. Flowchart of the BPME. The input data consist of degraded optical imagery from polar regions. The zero-reference metric system calculates the quantitative characterization of image degradation (Section III-A). Based on these metric vectors, BPM-P is generated through clustering mapping (Section III-B). Under the guidance of BPM-P, a selective enhancement is performed, leading to the fine reconstruction and output of the enhanced images (Section III-C).

correction [45], or discrete cosine transform coefficients [46]. Overall, feature-based methods possess a rigorous mathematical foundation and are well-suited for single images. The innovatively proposed intrinsic prior demonstrates considerable development potential in challenging polar image processing. Yet, given the unique characteristics of polar remote sensing imagery, existing mechanisms require further refinement.

III. Methodology

Fig. 2 shows the flowchart of the BPME, which follows a "perception-enhancement" paradigm and comprises three core modules.

A. Zero-Reference Metric System

To quantify the image degradation in the absence of external assistance, we first design a zero-reference metric system. It comprises three metrics with distinct physical significance that operate independently of rigid assumptions. Additionally, we introduce adaptive hyperparameters closely related to the image properties. This innovation circumvents the subjectivity and poor robustness of manual parameter tuning. This system can serve as a quantitative foundation for the subsequent generation of BPM-P.

1) Stochastic Noise Metric: Detection of stochastic noise is achieved through the metric P'. It is calculated by the inverse difference moment from the gray-level co-occurrence matrix. To align the P' with the image's dominant texture directions, we introduce an adaptive orientation θ_A based on the primary peaks $\{\theta_{\text{peak}}\}$ of the gradient orientation histogram. This design allows the metric to more accurately tell the difference between texture and noise. The metric P' is defined as follows:

$$P' = \sum_{(i,j)\in\Omega} \frac{P(i,j|\delta,\theta_A)}{1+(i-j)^2} \tag{1}$$

where $P(i,j|\delta,\theta_A)$ is the co-occurrence probability. A higher P' value indicates a highly homogeneous local region, while a lower value suggests significant gray-level variations.

2) Structural Information Metric: It is acknowledged that degraded texture manifests as deficient gradients, yet spatial continuity. To this end, the metric S' is formulated by (2). The process for computing S' incorporates adaptive hyperparameters θ_A and s_A. Specifically, the computation of θ_A aligns with that of P'. The s_A is set based on image resolution and the peak frequency of the power spectral density (PSD). This ensures that the metric corresponds to the physical scale of dominant structures

$$\begin{aligned} S' &= \sum_{s_A\in S}\sum_{\theta_A\in\Theta} w_{s_A,\theta_A}\left|C'_{s_A,\theta_A}(x,y)\right| \\ \text{s.t.}\quad C'_{s_A,\theta_A}(x,y) &= \mathcal{F}^{-1}\left(\hat{I}(\omega_x,\omega_y)\cdot G_{s_A,\theta_A}(\omega_x,\omega_y)\right) \end{aligned} \tag{2}$$

where $\hat{I}(\omega_x,\omega_y)$ is the frequency-domain representation of the image $I(x,y)$. $C'_{s_A,\theta_A}(x,y)$ and $G_{s_A,\theta_A}(\omega_x,\omega_y)$ are the subband coefficients and Gaussian filters, respectively. The weight w_{s_A,θ_A} reflects the normalized energy contribution of each subband.

3) Contrast Intensity Metric: The metric LC' is designed to quantify local luminance variations, thereby assessing contrast degradation. It is defined as the standard deviation within an adaptively sized window, referred to as W_A. Specifically, W_A

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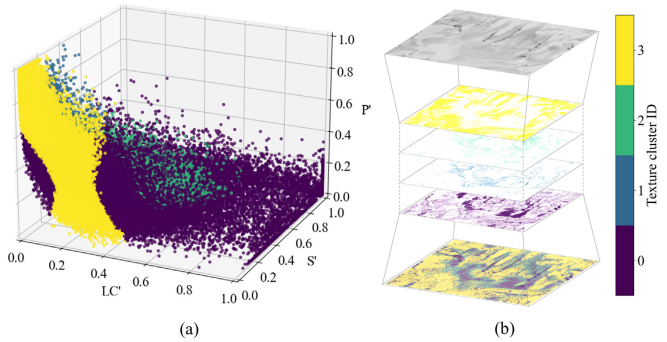


Fig. 3. Generation of the BPM-P. (a) Metric feature space. It intuitively illustrates the position and separability of the kth cluster, to demonstrate the descriptive reasonableness of the metric vectors. (b) Schematic illustration of BPM-P. Prior is acquired by discerning statistically significant patterns within the metric feature space via spatial mapping.

is proportional to the local correlation length derived from the image's autocorrelation function (ACF). This configuration ensures that the W_A are consistent with the scale of texture features. The following equation can be used to express LC':

$$LC' = \sqrt{\frac{1}{|W_A|}\sum_{(x,y)\in W_A}\left(I(x,y)-\mu_{W_A}\right)^2}$$
$$\text{s.t.} \quad \mu_{W_A} = \frac{1}{|W_A|}\sum_{(x,y)\in W_A} I(x,y) \tag{3}$$

where $|W_A|$ is the number of pixels and μ_{W_A} is the mean intensity within the W_A.

B. Generation of the BPM-P

1) Overview of BPM-P: Zero-reference metrics are designed for measuring features of degradation, but they struggle to show how these factors interact. To get around this, we translate the pixel-wise quantitative results into a spatially explicit BPM-P. The generation of priors is well-suited for polar applications, where obtaining labeled data or GT is prohibitive.

2) Unsupervised Clustering for Perception: We employ a Gaussian mixture model to cluster the feature vectors $\mathbf{t} = [S'_{\text{norm}}, P'_{\text{norm}}, LC'_{\text{norm}}]$ for each pixel. The mathematical formulation of the distribution is given by the following equation:

$$\mathcal{N}(\mathbf{t}|\mu_\mathbf{k}, \mathbf{\Sigma_k}) = \frac{1}{(2\pi)^{d/2}\,|\mathbf{\Sigma_k}|^{1/2}} \times \exp\left(-\frac{1}{2}(\mathbf{t}-\mu_\mathbf{k})^T\,\mathbf{\Sigma_k}^{-1}(\mathbf{t}-\mu_\mathbf{k})\right) \tag{4}$$

where $\mathcal{N}(\mathbf{t}|\mu_\mathbf{k}, \mathbf{\Sigma_k})$ is a Gaussian distribution, with mean $\mu_\mathbf{k}$ and covariance $\mathbf{\Sigma_k}$. $d = 3$, which represents the dimension of the metric feature space.

As shown in Fig. 3(a), to ensure objectivity and reproducibility, we utilize the Bayesian information criterion to automatically determine the optimal number of clusters, avoiding subjective selection. When the metric vector $\mathbf{t}$ follows a mixture model of k Gaussian distributions, its probability density function is as follows:

$$p(\mathbf{t}) = \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{t}|\mu_\mathbf{k}, \mathbf{\Sigma_k}) \tag{5}$$

where π_k is the weight of the kth Gaussian component, satisfying $\sum_{k=1}^{K}\pi_k = 1$.

For each pixel's metric feature vector $\mathbf{t}_i$, the following equation calculates its posterior probability, which belongs to the kth cluster:

$$p(k|\mathbf{t}_i) = \frac{\pi_k \mathcal{N}(\mathbf{t}_i|\mu_\mathbf{k}, \mathbf{\Sigma_k})}{\sum_{j=1}^{K}\pi_j \mathcal{N}\left(\mathbf{t}_i|\mu_j, \mathbf{\Sigma_j}\right)}. \tag{6}$$

Then, the cluster label l_i for this pixel is determined by the maximum posterior probability, and is given by the following formula:

$$l_i = \arg\max_k p(k|\mathbf{t}_i). \tag{7}$$

3) Spatial Visualization and Prior Generation: As depicted in Fig. 3(b), by mapping pixel-level cluster labels back to their spatial locations, we can analyze feature distributions to produce the BPM-P. This prior information effectively merges metric features with their spatial profiles. The mathematical expression of BPM-P is as follows:

$$\text{BPM-P} = \left\{\left(\mu_\mathbf{k}, \mathbf{\Sigma_k}, \pi_k, \text{Proportion}_k\right)\right\}_{k=1}^{K} \tag{8}$$

where Proportion_k represents the pixel proportion, and $\mu_\mathbf{k} = [\mu_{k_{S'}}, \mu_{k_{P'}}, \mu_{k_{LC'}}]$ and $\mathbf{\Sigma_k}$ of each cluster characterize a texture pattern.

Focusing on the enhancement demands, an adaptive recognition for BPM-P based on feature distribution percentiles is designed, defining the target enhancement region M_{target} and the noise reference region M_{noise}. Specifically, M_{target} comprises pixels falling below the 60th percentile of S'_{norm} and the 75th percentile of P'_{norm}, ensuring the selection of degradations with recoverable information. In contrast, M_{noise} isolates noise-dominated pixels above the 90th percentile of P'_{norm}. The data-driven recognition automatically adapts to the feature variability of each polar image.

C. Prior-Guided Selective Enhancement

Guided by the proposed BPM-P, this section outlines a selective enhancement. The method operates in the steerable pyramid domain to process the image at multiple scales and orientations. It consists of two key components controlling the enhancement, specifically the statistically driven signal selection and adaptive gain control.

1) Steerable Pyramid Decomposition: We choose the steerable pyramid [47] to decompose images into a structural layer and various texture layers. As illustrated in Fig. 4, this process allows us to manipulate specific features, including different spatial frequencies and angles. To preserve the image's fundamental radiometric values, we enhance the texture subbands, leaving the structural layer untouched. The steerable pyramid decomposition formula is as follows:

$$C_{s,\theta}(x,y) = \mathcal{F}^{-1}\left\{\mathcal{F}\{I(x,y)\}\cdot H_{s,\theta}(\omega_x, \omega_y)\right\}$$

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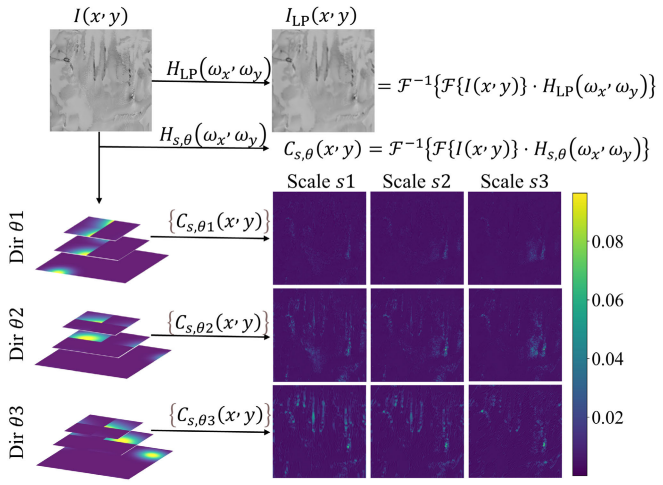


Fig. 4. Schematic illustration of steerable pyramid decomposition. The original image $I(x,y)$ is decomposed into a structural layer $I_{LP}(x,y)$ and texture layers. The texture layers, transformed into a coefficient-level feature domain, are represented by multichannel subband coefficients $\{C_{s,\theta}(x,y)\}$.

$$I_{LP}(x,y) = \mathcal{F}^{-1}\left\{\mathcal{F}\{I(x,y)\}\cdot H_{LP}(\omega_x,\omega_y)\right\}$$
$$I(x,y) \approx I_{LP}(x,y) + \sum_s\sum_\theta \mathrm{Re}\left\{C_{s,\theta}(x,y)\right\} \quad (9)$$

where $H_{s,\theta}(\omega_x,\omega_y)$ is the texture filter and $H_{LP}(\omega_x,\omega_y)$ is the structural filter. $\{C_{s,\theta}(x,y)\}$ and $I_{LP}(x,y)$ represent the subband coefficients and structural component after image decomposition. Re$\{\cdot\}$ denotes taking the real part.

To guarantee the decomposition domain's adaptability to the spatial heterogeneity of texture distribution in polar images, we design the texture filter $H_{s,\theta}(\omega_x,\omega_y)$ to be anisotropic, and its formula is as follows:

$$H_{s,\theta}(\omega_x,\omega_y) = \exp\left(-0.5\left[\left(\frac{\omega_\theta - f_s}{\sigma_{f_x}}\right)^2 + \left(\frac{\omega_{\perp\theta}}{\sigma_{f_y}}\right)^2\right]\right) \quad (10)$$

where ω_θ and $\omega_{\perp\theta}$, respectively, represent the frequency components in direction θ and its perpendicular direction. f_s represents the central frequency corresponding to scale s, σ_s is the spatial domain scale factor. $\sigma_{f_x} = 1/2\pi\sigma_s$ and $\sigma_{f_y} = 1/2\pi\gamma\sigma_s$ are frequency domain bandwidth control factors. γ adjusts the anisotropic effect, whose parameter settings are derived from θ_A.

Additionally, to ensure lossless information and minimal artifacts during processing, the structural filter $H_{LP}(\omega_x,\omega_y)$ is designed based on the principle of energy conservation. Its formula is given by the following equation:

$$H_{LP}(\omega_x,\omega_y) = \sqrt{\max\left(0, 1 - \sum_{s,\theta}|H_{s,\theta}(\omega_x,\omega_y)|^2\right)}. \quad (11)$$

2) Selective Enhancement: To distinguish valuable texture from noise, both of which are high-frequency signals, we devise a two-stage selective enhancement strategy. First, a thresholding based on statistical significance is applied. Only coefficients whose magnitudes exceed the significance threshold $\tau_{C,s,\theta}$ are retained as candidate texture signals. The subband noise standard deviation $\sigma_{n,s,\theta}$ is robustly estimated from the coefficients corresponding to the M_{noise}, and its formula is as follows:

$$\sigma_{n,s,\theta} = \mathrm{MAD}\left\{|C_{s,\theta}(x,y)| \mid \forall (x,y) \in M_{noise}\right\} \quad (12)$$

where MAD$\{\cdot\}$ represents the median absolute deviation.

Given the estimated $\sigma_{n,s,\theta}$, the significance threshold $\tau_{C,s,\theta}$ is determined by the following equation:

$$\tau_{C,s,\theta} = k_C \cdot \sigma_{n,s,\theta} \quad (13)$$

where k_C serves as the coefficient threshold factor, controlling the strictness of the significance threshold. We set $k_C = 2$ based on the statistical "two-sigma" rule for Gaussian noise, which provides a confidence level of approximately 95% to distinguish significant texture signals from noise.

To avoid boundary ringing effects and detail loss caused by the hard threshold, we introduce the sigmoid function as a smoothing activation function, and its formula is as follows:

$$A\left(|C_{s,\theta}(x,y)|\right) = \frac{1}{1 + e^{-\alpha_{sig}\left(|C_{s,\theta}(x,y)| - \tau_{C,s,\theta}\right)}} \quad (14)$$

where α_{sig} is the parameter that controls the transition bandwidth of the sigmoid function. $\alpha_{sig} \approx 2/\sigma_{n,s,\theta}$ is determined based on the significance threshold design. This step effectively suppresses noise interference while maximizing the preservation of texture information.

Second, for the coefficients retained within M_{target}, an adaptive gain $G_{BPME,s,\theta}$ is applied for enhancement. This gain is inversely proportional to the mean deviation of contrast intensity $\Delta\overline{LC'_{M_{target}}}(x,y)$ relative to M_{target}, preventing over-enhancement in areas that already possess sufficient contrast. The enhanced coefficient $C'_{s,\theta}(x,y)$ is computed as follows:

$$G_{boost}(x,y) = G_{base} \cdot \max\left(1, \left(1 - \Delta\overline{LC'_{M_{target}}}(x,y)\right)^{\lambda_{LC'}}\right)$$
$$G_{BPME,s,\theta}(x,y) = 1 + M_{target}(x,y) \cdot G_{boost} \cdot A \cdot w_{s,\theta}$$
$$C'_{s,\theta}(x,y) = C_{s,\theta}(x,y) \cdot G_{BPME,s,\theta}(x,y) \quad (15)$$

where $w_{s,\theta}$, based on energy normalization, is the weight factor for the subband $C_{s,\theta}(x,y)$. G_{base} represents the base gain parameter, controlling the overall magnitude of enhancement. $\lambda_{LC'}$ is the gain intensity control parameter. Based on the analysis of robustness and performance trade-offs, we configure $\lambda_{LC'} = 1$ and $G_{base} = 1.5$ to establish a stable negative feedback, ensuring substantial enhancement while strictly preserving the image's structural integrity.

3) Enhanced Image Reconstruction: The enhanced image is reconstructed by applying the inverse steerable pyramid transform to the $I_{LP}(x,y)$ and the $\{C'_{s,\theta}(x,y)\}$. The formula is as follows:

$$I_{BPME}(x,y) = \mathrm{Re}\left\{I_{LP}(x,y) + \sum_{s,\theta} C'_{s,\theta}(x,y)\right\}. \quad (16)$$

IV. Experiments

We conducted a series of experiments in this section to evaluate BPME. In particular, Section IV-A initially presents the details of experimental settings. Section IV-B assesses the effectiveness of our proposed BPM-P. An extensive comparison of BPME with other cutting-edge algorithms across

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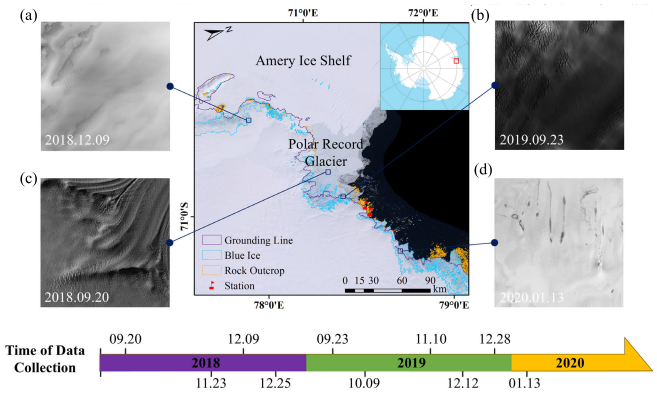


Fig. 5. Study area of the experimental data. The central map shows the distribution of stations, grounding line [48], blue ice [49], and rock outcrop [50] within the geographic coverage. The sub-images on both sides show the image examples of the four categories of data, with the acquisition time marked in the lower left corner. The upper part of them illustrates the corresponding category ID and image characteristics. In parentheses, the data volume of each category is given. In addition, the time axis at the bottom shows the imaging times of the experimental data. (a) Inherent weak texture (39). (b) Non-ideal meteorology (35). (c) Complex illumination (34). (d) Ideal texture (32).

various degradation scenarios is offered in Section IV-C. Section IV-D ultimately evaluates the practical application of our method in the downstream tasks.

A. Experimental Settings

1) Datasets: To adequately evaluate the performance of BPME, experimental data were collected under challenging polar imaging conditions. The original data were obtained from the 15-m panchromatic bands of the Landsat 8 OLI/TIRS Level-1 products. The study area is the eastern front of the Amery Ice Shelf, which is located in East Antarctica. It encompasses vast snowfields, blue ice zones, and ice crevasses of various scales, providing a representation of the diversity of polar texture features. Acquired across two full melt seasons (2018–2020), the images comprise a range of degradation scenarios, such as severe illumination variations and meteorological interference. Considering the efficiency of experiments, the original images were cut into non-overlapping 512×512 pixel patches. This scale retains details of local texture while capturing geomorphological aspects of the region. Additionally, the experimental data were divided into four main categories of texture features and degradation conditions. As shown in Fig. 5, the data comprise 140 representative image patches with balanced class distribution, ensuring fair performance evaluation.

2) Implementation Details: We compared our proposed method with classical and state-of-the-art techniques from the three paradigms reviewed in Section II. When selecting classical methods, we prioritized those that had been maturely applied to polar image processing to ensure the validity of the evaluation. To maintain objectivity, all comparison methods strictly adhered to the configuration parameters recommended in the original literature. For deep learning methods, we used pretrained models provided by the developers. All experiments were conducted under standardized conditions consisting of the NVIDIA GeForce RTX 4050 GPU and Intel Core i5-13500H CPU, using the PyTorch 2.2.1 framework with CUDA 11.8.

3) Evaluation Metrics: The evaluation employed a combination of qualitative and quantitative methodologies. The selection of metrics considered the unique characteristics of polar imaging environments and consistency with human visual perception. The quantitative evaluation design utilized nonreference metrics as its core, with performance assessment dimensions focused on key factors impacting downstream application value. The multidimensional evaluation framework incorporated physical fidelity metrics, such as patch-based contrast quality index (PCQI) [51] and noise quality measure (NQM) [52]; texture quality metrics, including "texturedness" [53], [54] and wavelet high-frequency energy (E_{WHF}); the contrast enhancement metric, enhanced modified edge (EME) [55]; and the information entropy metric, Shannon entropy [56] (H_{SE}). The E_{WHF} is calculated as follows:

$$E_{\mathrm{WHF}} = \sum_{l=1}^{L} \left(\sum_{m,n} \left| D_l^H(m,n) \right| + \sum_{m,n} \left| D_l^V(m,n) \right| + \sum_{m,n} \left| D_l^D(m,n) \right| \right) \tag{17}$$

where $|D_l^H(m,n)|$, $|D_l^V(m,n)|$, and $|D_l^D(m,n)|$ are the horizontal, vertical, and diagonal detail coefficients from the wavelet decomposition of the $M \times N$ image.

Notably, NQM belongs to the "cost" metric $\mathcal{M}_{\mathrm{cost}}$, where lower values indicate reduced noise distortion, while the other metrics are the "benefit" metrics $\mathcal{M}_{\mathrm{benefit}}$, with higher values reflecting superior image quality. To facilitate a comparison across these heterogeneous dimensions, we further apply the Min-Max normalization to define an "Overall Performance Score" for each image. The derivation follows an equal weighting scheme, ensuring an unbiased evaluation.

$$\begin{aligned} &\text{Overall Performance Score}_k \\ &\quad = \frac{1}{N} \sum_{j=1}^{N} s_{k,j} \\ &\text{s.t.} \quad s_{k,j} = \begin{cases} \dfrac{v_{k,j} - v_{j,\min}}{v_{j,\max} - v_{j,\min}}, & j \in \mathcal{M}_{\mathrm{benefit}} \\ \dfrac{v_{j,\max} - v_{k,j}}{v_{j,\max} - v_{j,\min}}, & j \in \mathcal{M}_{\mathrm{cost}} \end{cases} \end{aligned} \tag{18}$$

where $v_{k,j}$ refers to the raw value of the jth metric for the kth algorithm. $v_{j,\max}$ and $v_{j,\min}$ refer to the extremum values across all comparative algorithms for the current image. $s_{k,j} \in [0,1]$ denotes the normalized result.

B. Evaluation of the BPM-P

This section systematically analyzes the three feature metrics that are built into BPM-P. We thoroughly assess their perceptual performance in cases of polar image degradation. Through rigorous comparison with widely recognized existing evaluation metrics, we further demonstrate their validity and robustness under extreme polar conditions.

As shown in Fig. 6, under ideal conditions, the proposed P' exhibits quasi-binarization properties. Its low-response regions

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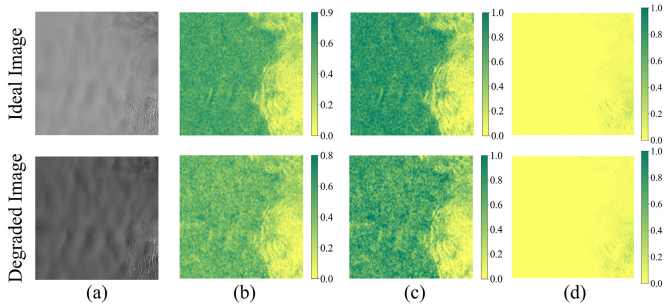


Fig. 6. Visualization of metrics in the noise evaluation dimension. (a) Original image. (b) P' visualization. (c) and (d) Normalized results of P' and the Laplacian operator. Due to differing evaluation principles, the low-value regions of P' in the visualization correspond to the components represented by the high-value responses of the Laplacian operator.

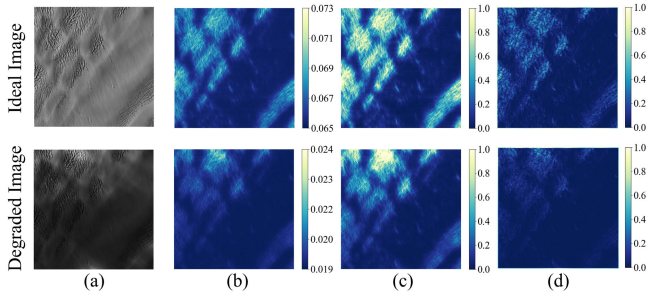


Fig. 7. Visualization of metrics in the structure evaluation dimension. (a) Original image. (b) S' visualization. (c) and (d) Normalized results of S' and the Sobel operator.

accurately reflect high-frequency content, analogous to the high-value performance of the noise-sensitive Laplacian operator [57] when extracting such features. However, standard high-frequency response operators do not work as well in low signal-to-noise ratio scenarios, which are common in degraded images. To address this problem, P' introduces adaptive directional factors, which make it easier to detect and describe stochastic noise. By accurately identifying discrete low-value points and indistinct boundaries, P' provides a quantitative tool for assessing noise degradation, overcoming the constraints of fixed-response methodologies.

The efficacy of S' in structural information evaluation is illustrated in Fig. 7. S' shows resistance to noise in both ideal and inadequate images, which makes it possible to accurately describe how structural information is spread out. Its performance closely corresponds to that of the classical Sobel operator [58]. However, unlike conventional gradient-aware operators, S' effectively integrates adaptive scale and directional factors. Consequently, even amidst structural degradation and weak textures, S' yields substantially more robust and finely resolved assessments compared to operators solely reliant on gradient-based quantification.

Fig. 8 demonstrates the performance of LC'. The contrast characterization of LC' demonstrates a robust association with the global standard deviation contrast evaluation metric, ensuring comprehensive consistency. However, its primary strength lies in addressing localized degradation. LC' employs an adaptive window that changes based on the texture scales. Unlike methods that depend on global parameters, this attribute makes

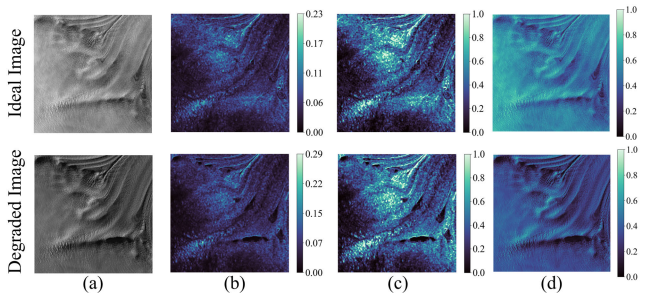


Fig. 8. Visualization of metrics in the contrast evaluation dimension. (a) Original image. (b) LC' visualization. (c) and (d) Normalized results of LC' and the global contrast metric.

it possible to accurately analyze contrast in challenging areas, such as topographic shadows.

C. Comparison of Enhancement Algorithms

1) Qualitative Analysis: Qualitative assessment was primarily conducted via visual inspection, further aided by image histograms and absolute gain heatmaps to facilitate comparative analysis.

a) Comparison with physical model-based methods: As shown in Fig. 9, DCP [23] struggles with surfaces like ice, where its core assumption is liable to failure. This leads to the histogram being crammed into the low-gradient range, ultimately losing significant detail. The SLP [25] is more versatile because it looks at local pixel saturation. The problem is that unique landscapes in polar regions can disrupt the model, leading to underperformance. The Retinex-based models, such as LIME [26] and DUAL [27], are primarily designed to improve low-light scenes. LIME often estimates the lighting incorrectly, which results in overexposed bright spots. And while DUAL offers some improvement, its model can't really handle the varied interferences of polar imagery, producing unsatisfactory histograms with multiple peaks. In contrast, BPME's gain heatmaps show it targets low-texture regions to boost gain. And the continuous histograms it produces prove how robust it truly is.

b) Comparison with zero-reference deep learning networks: Fig. 10 compares our method to zero-reference deep learning methods. Because the design of its network relies on the DCP assumption, ZID [28] inevitably performs unrealistic distortions. Zero-DCE [31] and ZERO-IG [35] can boost overall brightness, but their loss functions do not generalize well in polar imagery. Such behavior often leads to amplified noise and blown-out highlights. Their histograms are sharp spikes or clipped high-value regions. SGZ [38] takes a different route, using high-level semantic information to achieve more natural results. But since it was trained on low-light datasets, it struggles to meet the weak texture enhancement in polar scenarios. In comparison, BPME is highly spatially adaptive. The absolute gain heatmaps confirm that it focuses on textures for enhancement. The much smoother histogram distributions prove that it avoids amplifying noise and overexposing the image.

c) Comparison with feature-based methods: Fig. 11 shows the results against feature-based methods.

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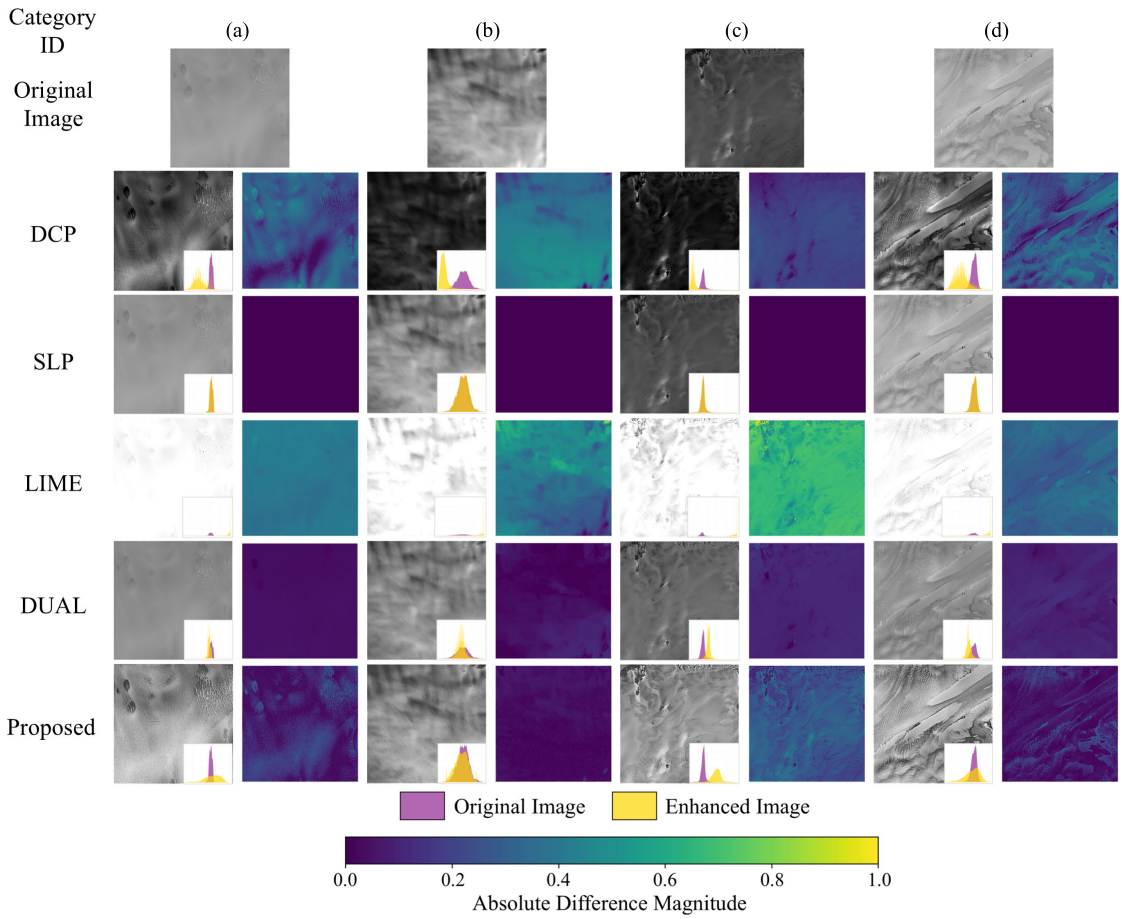


Fig. 9. Visual comparison of physical model-based methods. In each experimental category, the first column illustrates the visualization outcomes of various methods, including the histogram distribution of original and enhanced images at the bottom-right. The second column depicts the corresponding absolute gain heatmap. (a) Inherent weak texture. (b) Non-ideal meteorology. (c) Complex illumination. (d) Ideal texture.

Histogram-based methods like CLAHE enhance the contrast uniformly. This brute-force approach tends to create visual artifacts and warp the natural pixel distribution of the image. The Wallis (WALLIS) filter [40] amplifies noise in flat areas because it is difficult to tell the veritable texture apart from noise. The same flaw is in high-frequency filters like highpass filter (HIGHPASS) [41], Sobel (SOBEL) [42], and Laplacian (LAP) [59]. This limitation invariably leads to their histograms exhibiting a constricted range and pronounced peaks. The NAOF [43] algorithm obtains abundant textures, but polarizes grayscale values into a "U"-shaped histogram, leading to a loss of midtones. Furthermore, the gain heatmaps above show that their techniques for improvement mostly lack spatial adaptability. Concurrently, the BPME conducts a mechanism that perceptually improves texture regions, making it easy to adjust to degradation that varies by space. The histogram distribution display further supports the idea that our method improves image quality and reduces noise at the same time.

2) Quantitative Comparison: Qualitative analysis indicated that the dark channel model added intensity aberrations, whereas CLAHE significantly damaged image details. Because of the limitations, they were not included in the quantitative analysis to make sure the evaluation results made sense. With a focus on texture enhancement and detail preservation, which are the key technical requirements of polar image processing, the remaining algorithms were evaluated based on three aspects.

a) Physical fidelity evaluation: The core challenge of texture enhancement algorithms lies in effectively improving the discriminability while maintaining the macroscopic structure. To quantitatively evaluate this performance, a 2-D system is established, where texturedness quantifies the enhancement of texture, and PCQI assesses the fidelity to the image's low-frequency components. The average values of metrics are shown in Fig. 12(a) as a scatter plot. This graph illustrates the performance between texture enhancement and physical fidelity for different algorithms.

As seen in Fig. 12(a), the NAOF produced the highest value of texturedness, a superior result of its anisotropic filtering. However, the trade-off is a very low PCQI value, which means it likely distorts the structure when it sharpens textures. The WALLIS ran into a similar problem. While it boosted the texturedness, its PCQI values stayed below 0.8. By comparison, BPME was the clear standout. Its scatter points consistently land near the ideal upper-right corner of the plot. The PCQI values remained above 0.95, demonstrating its exceptional ability to maintain significant physical characteristics. At the same time, its texturedness values were far better than the rest of the methods. All the evidence indicates that BPME strikes

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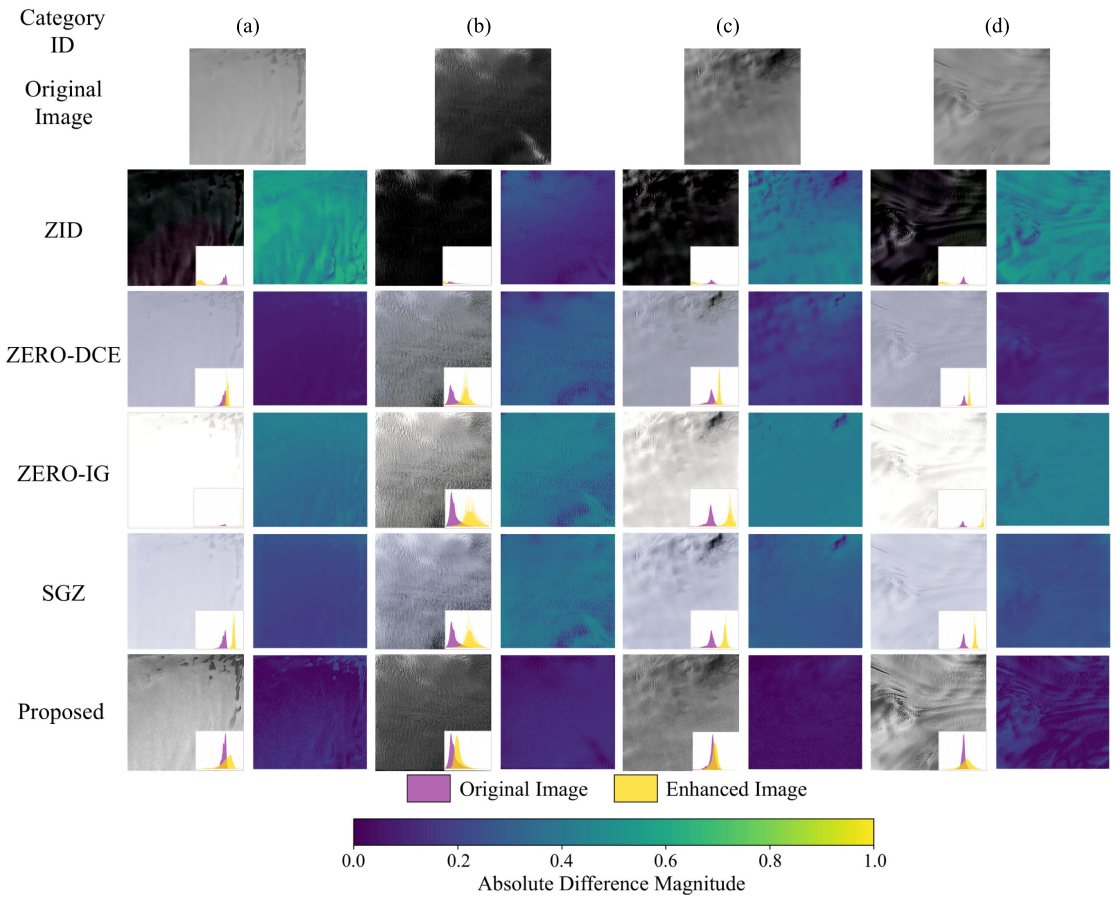


Fig. 10. Visual comparison of zero-reference deep learning networks. (a) Inherent weak texture. (b) Non-ideal meteorology. (c) Complex illumination. (d) Ideal texture.

the right balance, enhancing weak textures without sacrificing the physical accuracy of the images.

b) Noise suppression evaluation: For quantitatively assessing each algorithm's robustness against noise amplification during enhancement, E_{WHF} and the NQM are employed. E_{WHF} quantifies the high-frequency energy, while NQM measures additional noise added relative to the original image. Fig. 12(b) presents the quantitative evaluation results.

Algorithms such as NAOF, WALLIS, LAP, and SOBEL are noteworthy in sharpening E_{WHF}. When faced with both texture and stochastic noise, which are both part of the high-frequency energy, their capacity to identify enhanced objects is inadequate in the degraded situation, nevertheless. As a result, they inevitably achieve the NQM with high values. LIME and Zero-DCE excel in complex illumination data, on account of being built specifically for low-light degraded scenes. But they stumbled when faced with other kinds of degradation, indicating a limited capacity for generalization. Meanwhile, most of the other algorithms' metrics are almost the same as the original image, which suggests that they fail to improve degradation in any meaningful way. BPME delivers a substantial boost to the E_{WHF} scores while only causing a small bump in the NQM values. This outstanding performance shows it can intelligently pick out texture to enhance while actively suppressing noise.

c) Information richness evaluation: The contribution of enhancement algorithms to information richness was quantitatively evaluated using EME and H_{SE}. EME assesses local contrast, while H_{SE} quantifies the richness of image information. Fig. 12(c) illustrates the scatter distribution of their mean values.

The WALLIS and NAOF produce high H_{SE} values. Nonetheless, being informed of the inconsistent EME values, the boost in H_{SE} tends to come from artificial randomness, not from any genuine information in the image. This process creates artifacts that show up as "pseudo-textures" in otherwise smooth regions. On the flip side, HIGHPASS and LAP performed well on the EME metric but poorly on H_{SE}. This happens because they are designed to sharpen edges by enhancing high-frequency components. While this type of enhancement is great for local contrast, it comes at a cost, as the low-frequency components that hold global information are sacrificed. BPME achieves a much better balance. It delivers a solid improvement in EME while also earning a high H_{SE} score. This indicates that BPME can enhance local contrast without losing the broader information content of the images.

d) Comprehensive capability analysis and ranking: Based on the overall performance score defined in (18), this section quantified the comprehensive capabilities of all algorithms. Table I details the average results from the experimental data. The results indicate that the BPME algorithm obtained the highest score, demonstrating a superior balance across the three aforementioned evaluation aspects. Its average

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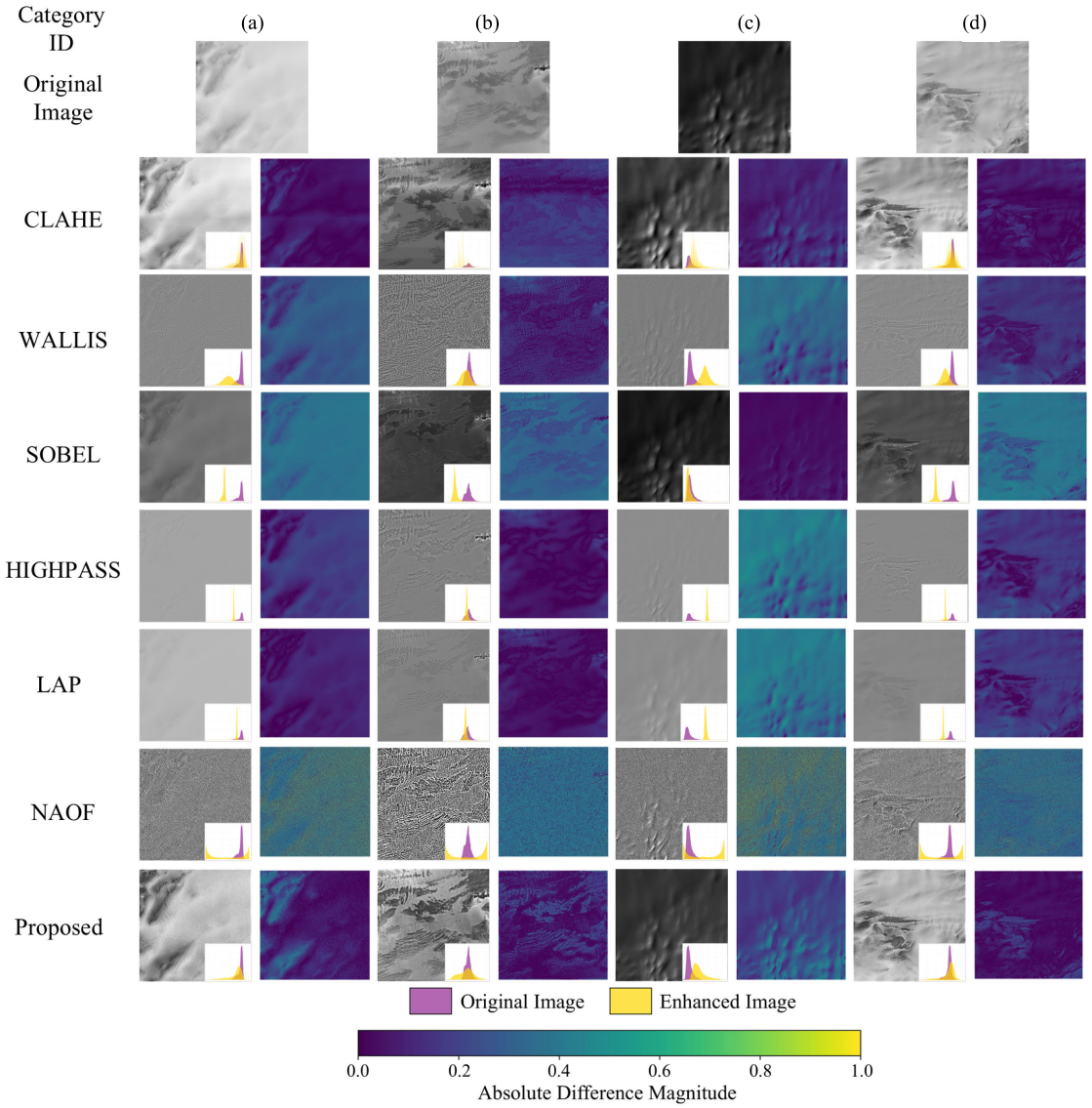


Fig. 11. Visual comparison of feature-based methods. (a) Inherent weak texture. (b) Non-ideal meteorology. (c) Complex illumination. (d) Ideal texture.

TABLE I
OVERALL PERFORMANCE SCORES ACROSS VARIOUS ENHANCEMENT ALGORITHMS

Algorithm	LIME	DUAL	Zero-DCE	ZERO-IG	SGZ	Proposed
Overall Performance Score	0.265	0.395	0.371	0.284	0.380	**0.568**
Algorithm	WALLIS	SOBEL	HIGHPASS	LAP	NAOF	
Overall Performance Score	0.354	0.357	0.338	0.319	0.518	

score reached 0.568, which led the second-place method by 9.65%.

D. Validation of Downstream Task Performance

To validate the practical value of BPME, we designed a verification process involving typical downstream tasks. The NAOF and HIGHPASS were identified as the most competitive for comparative experiments in these tasks based on comprehensive capability rankings. The effectiveness of these algorithms in enhancing contours and pixel-level features was thoroughly evaluated.

1) Contour Extraction Performance Analysis: As illustrated in Fig. 13, the classic Canny operator [60] was employed for contour feature extraction. The NAOF yielded the most abundant contour features. However, these extracted features appeared chaotic in the presence of amplified noise, which limited their interpretability. The HIGHPASS focuses on high-frequency signal processing but lacks the capability to sufficiently enhance weak texture signals. Its gain effect for contour extraction tasks is suboptimal. In contrast, the BPME algorithm produces the clearest and most continuous contour features. The improvement is crucial for subsequent

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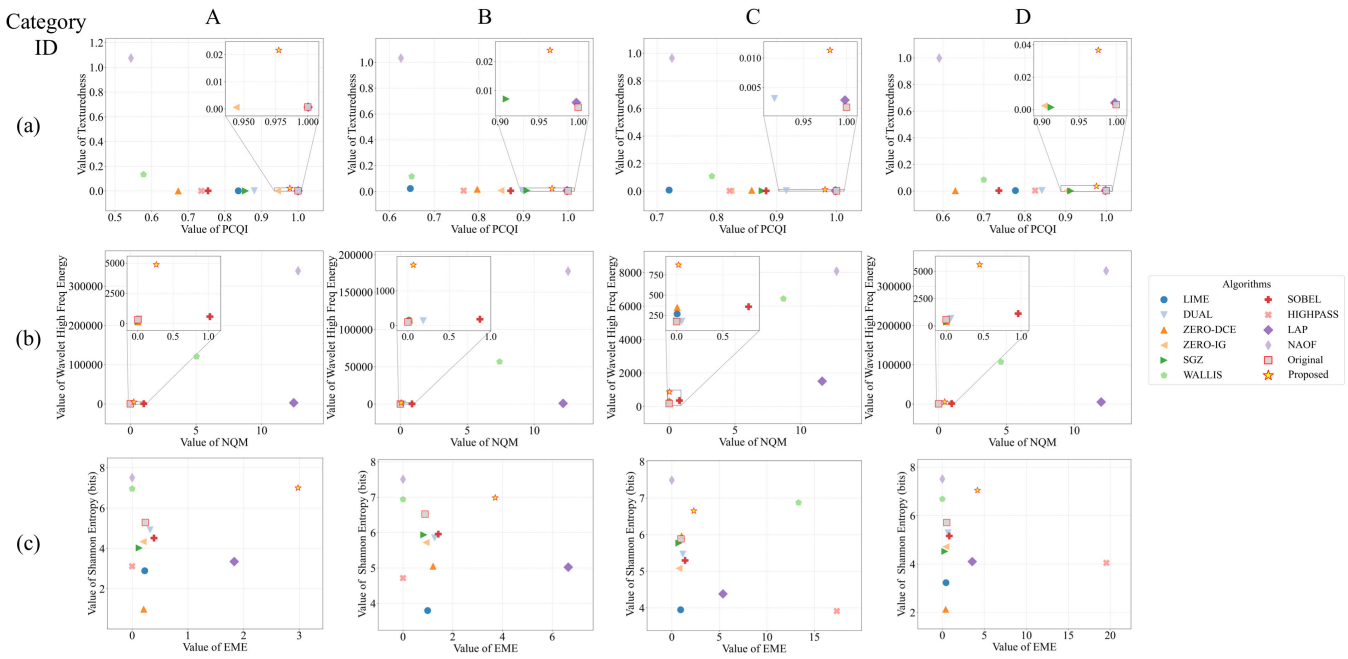


Fig. 12. Scatter plot of quantitative evaluation metrics for various algorithms. A magnified local view is provided in the upper-right or upper-left corner. (a) Physical fidelity evaluation. An optimal algorithm should maximize texturedness and PCQI. (b) Noise suppression evaluation. The increase in NQM should be reduced by an ideal algorithm while E_{WHF} is maximized. (c) Information richness evaluation. A superior algorithm should increase H_{SE} and EME.

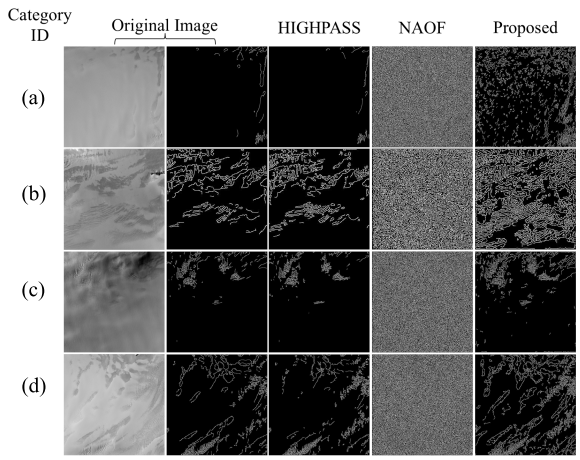


Fig. 13. Visual comparison of contour extraction performances from enhancement algorithms. (a) Inherent weak texture. (b) Non-ideal meteorology. (c) Complex illumination. (d) Ideal texture.

geometric modeling and analysis, particularly in challenging polar regions, where ideal information is difficult to obtain.

2) Ice Flow Velocity Tracking Performance Evaluation: To objectively validate the effectiveness of the proposed algorithm's enhancement on ice flow velocity tracking performance, the advanced algorithm RoMa [61] was employed for pixel-level ice flow velocity tracking. To ensure the reliability of this performance assessment, the MEaSUREs project InSAR velocity products [62], [63] served as the reference for tracking direction and magnitude. The resulting tracking performance is presented in Fig. 14.

Fig. 14 shows that original degraded images often made tracking challenging, which often resulted in data gaps or incorrect matches. Enhancement methods effectively ameliorated tracking performance in most challenging regions,

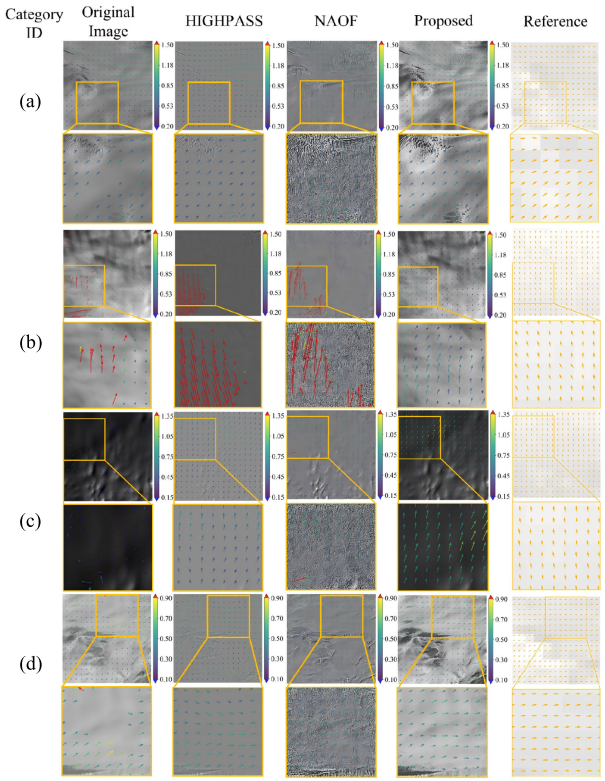


Fig. 14. Visual comparison of winsorized clipping colormap for ice flow velocity tracking performances, with magnified local views. The color bar indicates the magnitude of velocity, measured in meters per day. The MEaSUREs project InSAR velocity products [62], [63] served as the reference. The MOA product [64] was utilized as the base map, with tracking vector fields visualized at a 30-pixel grid sampling interval. (a) Inherent weak texture. (b) Non-ideal meteorology. (c) Complex illumination. (d) Ideal texture.

thereby confirming their pivotal role in ensuring robust downstream task quality. However, non-ideal meteorology data,

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which had a lot of structural noise from fog occlusion, was a more difficult scene that clearly showed the limitation of comparative algorithms. Specifically, HIGHPASS struggled because it was unable to suppress strong noise interference, causing major distortions that led to widespread tracking failures. NAOF ran into trouble because it was good at handling the isotropic noise, so it was ineffective against the structural kind. However, BPME's selective enhancement was very robust against significant structural noise and always produced smooth and natural displacement fields.

V. Conclusion

Optical remote sensing imagery is crucial for polar monitoring, yet its utility is severely limited by significant texture degradation in polar regions. To address this, we propose the interpretable BPME, which enables adaptive enhancement without relying on auxiliary data. Comprehensive experiments indicate that BPME offers considerable benefits in multidimensional evaluation and downstream tasks. On account of its enhancement mechanism relies on intrinsic properties rather than sensor-specific parameters or spectral response functions, and the study area's diverse topography and complex atmospheric conditions effectively represent polar imaging scenarios. BPME holds significant potential to be extrapolated to other single-band optical imagery.

However, due to the BPM-P that depends on retrievable information, it may be ineffective in scenarios, such as thick cloud occlusion. And processing bands independently may disrupt spectral fidelity for multispectral data. Our future work will leverage the advantages of deep learning to further enhance algorithmic intelligence, while focusing on incorporating interband constraints into metric-driven optimization.

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